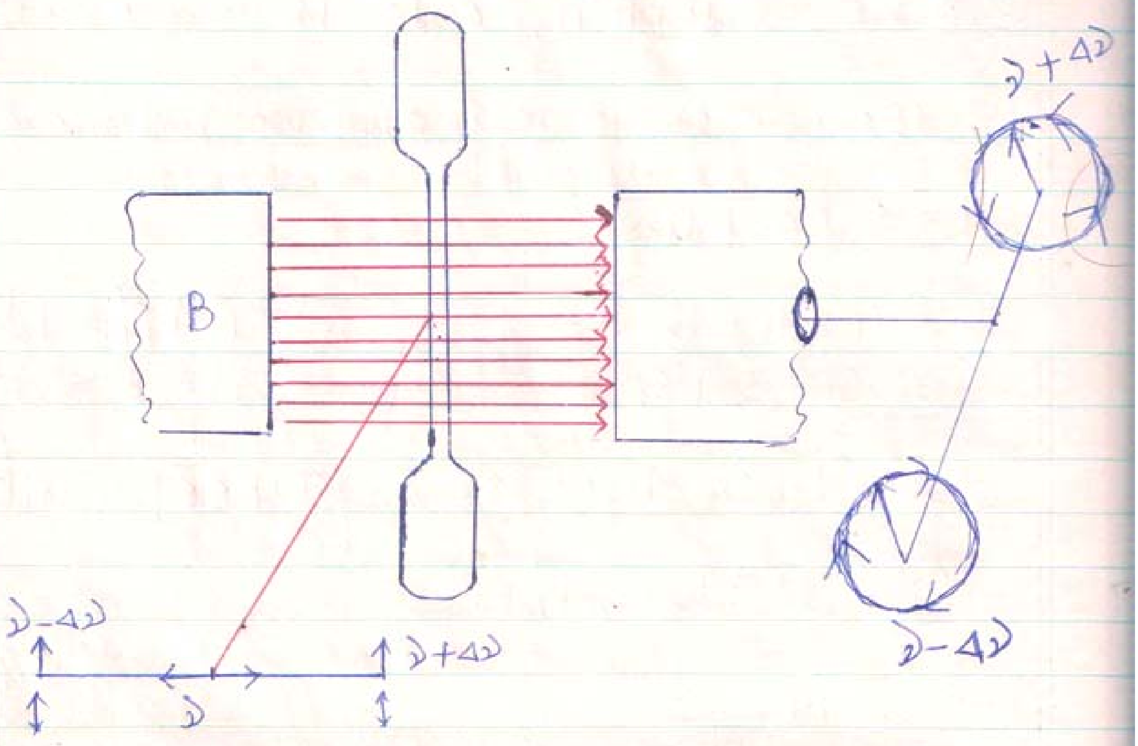
Zeeman Effect:



When a source of light is kept in a strong uniform magnetic field; it is found that looking from a direction $\perp^r$ to the applied field; each spectral line of frequency 'ν' splits into three lines of frequency $(\nu - \Delta\nu)$, ν & $(\nu + \Delta\nu)$ and looking from a direction along the direction of the magnetic field each line of frequency ν splits into two lines of frequency $(\nu - \Delta\nu)$ & $(\nu + \Delta\nu)$. This phenomenon of splitting of spectral lines in the presence of a strong magnetic field is known as normal Zeeman effect.

In transverse Zeeman effect the spectrometer is set at right angles to the direction of applied

$\lambda_1 = 4896\ A^\circ$ $D_1 =$

$\lambda_2 = 4890\ A^\circ$ $D_2 =$

magnetic field and each spectral lines splits into a triplet. All the three lines are found to be plane polarised. The line of unchanged frequency ν has vibration along the direction of applied field, while the other two lines has frequency $(\nu - \Delta\nu)$ & $(\nu + \Delta\nu)$ have vibrations along perpendicular to the direction of the magnetic field.

In normal longitudinal Zeeman effect; the spectrometer is set along the direction of magnetic field through bored pole pieces. Each line splits into a doublet of frequency $(\nu - \Delta\nu)$ & $(\nu + \Delta\nu)$ both of which are circularly polarised light; having rotations in opposite directions.

However the splitting of a single line into a triplet is by far less common in the spectrum of elements. The spectral lines of most of the elements when subjected to a strong external field; splits into a multiplet and the number of lines depends on the state of transition and this phenomenon is known as Anomalous Zeeman Effect.

The D_1 lines in Na, splits into four lines while D_2 splits into six lines in presence of a strong magnetic field.

Explanation:

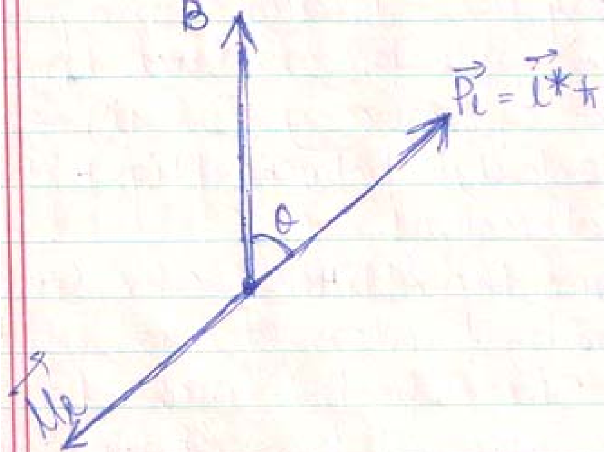
Normal Zeeman Effect: Normal Zeeman Effect is observed in atoms having total spin Quantum number $S = 0$.

The orbital angular momentum of the electron $= \vec{P_l} = \vec{l^*}\hbar$, $|\vec{P_l}| = l^*\hbar$

$\vec{B}$ = The induction vector of the applied magnetic field.

$\vec{\mu_l}$ = The magnetic moment due to the orbital motion of the electron $= \frac{e}{2m} \vec{P_l} = \frac{e\hbar l^*}{2m}$.

Let θ = The angle between the direction of orbital angular momentum vector $\vec{P_l}$ (or $\vec{\mu_l}$) and the direction of the applied magnetic field $\vec{B}$.



$C = MB\sin\theta$
$= M \times B$

The torque or couple experienced by the electron due to the interaction between the applied magnetic field $\vec{B}$ & the orbital magnetic moment $\vec{\mu_l}$ is $\vec{C} = \vec{\mu_l} \times \vec{B}$ —— (1)

$$|\vec{C}| = C = \mu_l B \sin\theta = \frac{e\hbar l^*}{2m} . B . \sin\theta \quad \text{—— (2)}$$

This torque or couple produces a change in angular momentum of the electron; given by

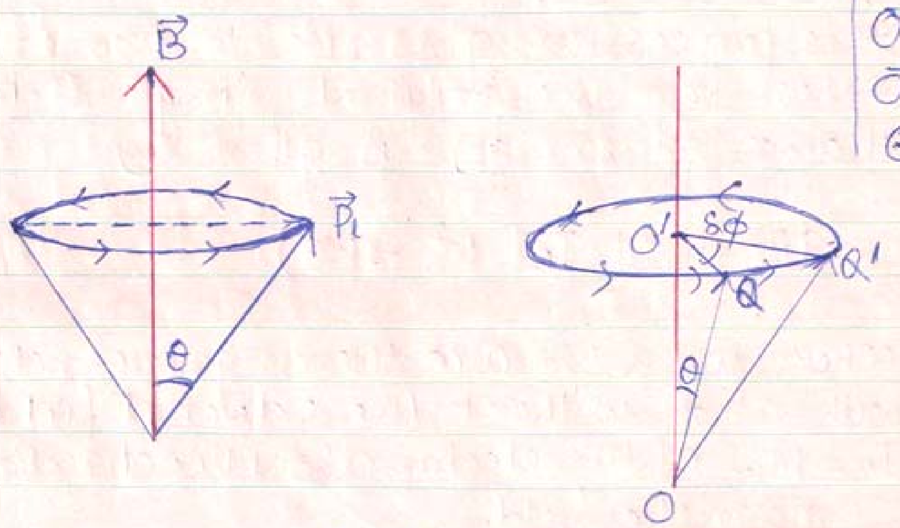
$\vec{C}$ = Rate of change of momentum.

$$\vec{C} = \frac{d(\vec{P_l})}{dt} \quad \text{or} \quad C = \frac{d(l^*\hbar)}{dt} \quad \text{—— (3)}$$

By the rule of cross-product ; $\vec{C}$ is $\perp^r$ to both $\vec{\mu_l}$ & $\vec{B}$. Since the direction of the couple is $\perp^r$ to the plane of μ_l (i.e $\vec{P_l}$) & $\vec{B}$ this couple instead of changing the magnitude of angular momentum, changes the direction of angular momentum.

As a result the angular momentum vector precesses around the direction of $\vec{B}$ at const. angular velocity, the tip of the vector describes a circle of radius $P_l \sin\theta$.

$\vec{OQ} = \vec{P_l}$
$\vec{OQ'} = \vec{P_{l'}}$
$\vec{QQ'} = \vec{P_{l'}} - \vec{P_l}$



let the vector $\vec{OQ}$ $(=\vec{P_l})$ go to $\vec{OQ'}$ $(=\vec{P_l})$ in time δt covering an angle $Q\hat{O'}Q' = \delta\phi$ at the centre.

Using definition of angle :- from $QO'Q'$; $\delta\phi = \frac{QQ'}{r}$

or $\delta\phi = \frac{\delta(P_l)}{P_l \sin\theta}$

The angular velocity of precession is

$$\omega = \lim_{\substack{\delta\phi \to 0 \\ \delta t \to 0}} \frac{\delta\phi}{\delta t} = \lim_{\substack{\delta P_l \to 0 \\ \delta t \to 0}} \left[\frac{\delta(P_l)}{(P_l \sin\theta)\delta t}\right] = \frac{1}{P_l \sin\theta}\frac{d(P_l)}{dt}$$

Putting equation ③: $\omega = \dfrac{C}{P_1 \sin\theta}$

Putting the values of C (from eq^n ②) & P_1

$$\omega = \frac{\dfrac{e\hbar l^* B\sin\theta}{2m}}{l^*\hbar\sin\theta} = \frac{eB}{2m}$$

$$\boxed{\omega = \frac{eB}{2m}} \quad \text{— ④}$$

This precession gives rise to two energy States from the original Single State. The change in energy is given by

$$\Delta E = \frac{1}{2} I_B(\omega_B^2 - \omega_0^2)$$

Where ω_B & ω_0 are the angular velocity with and without the external field.

I_B = M.I of the electron about the direction of magnetic field.

$$\therefore \Delta E = \frac{1}{2} I_B(\omega_B + \omega_0)(\omega_B - \omega_0)$$

But $\omega_B - \omega_0 = \omega = \dfrac{eB}{2m}$

& $\omega_0 \simeq \omega_B \quad \therefore \omega_B + \omega_0 \simeq 2\omega_0$.

$$\therefore \Delta E = \frac{1}{2} I_B . 2\omega_0 . \frac{eB}{2m} = \frac{I_B\omega_0 eB}{2m} \quad \text{— ⑤}$$

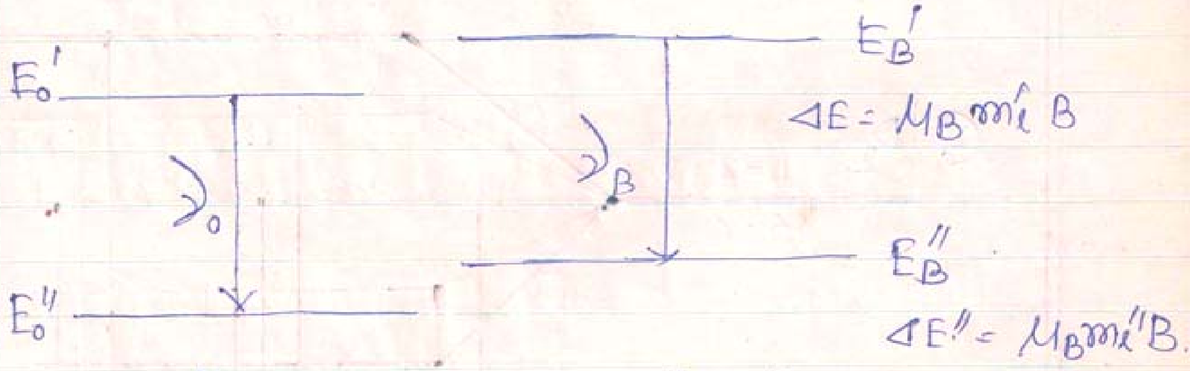
But $I_B \omega_0$ = The orbital angular momentum in the direction of the applied field. The Z-component of the orbital angular momentum $(P_l)_z = m_l \hbar$ where m_l = Orbital magnetic Quantum no. having $(2l+1)$ values from $-l$ to $+l$ in steps of unity.

$\therefore \Delta E = m_l \hbar \dfrac{eB}{2m}$ But $\dfrac{e\hbar}{2m} = \mu_B$ = Bohr Magneton

$$\therefore \boxed{\Delta E = \mu_B m_l B} \quad \text{—— (6)}.$$

let E_0' & E_0'' be the energy of the two excited States (between which electron jumps) before applying the magnetic field.

E_B' & E_B'' = The energy values when the magnetic field is applied.



$\therefore$ The frequency of the spectral lines when the electron jumps in the presence of magnetic field is given by

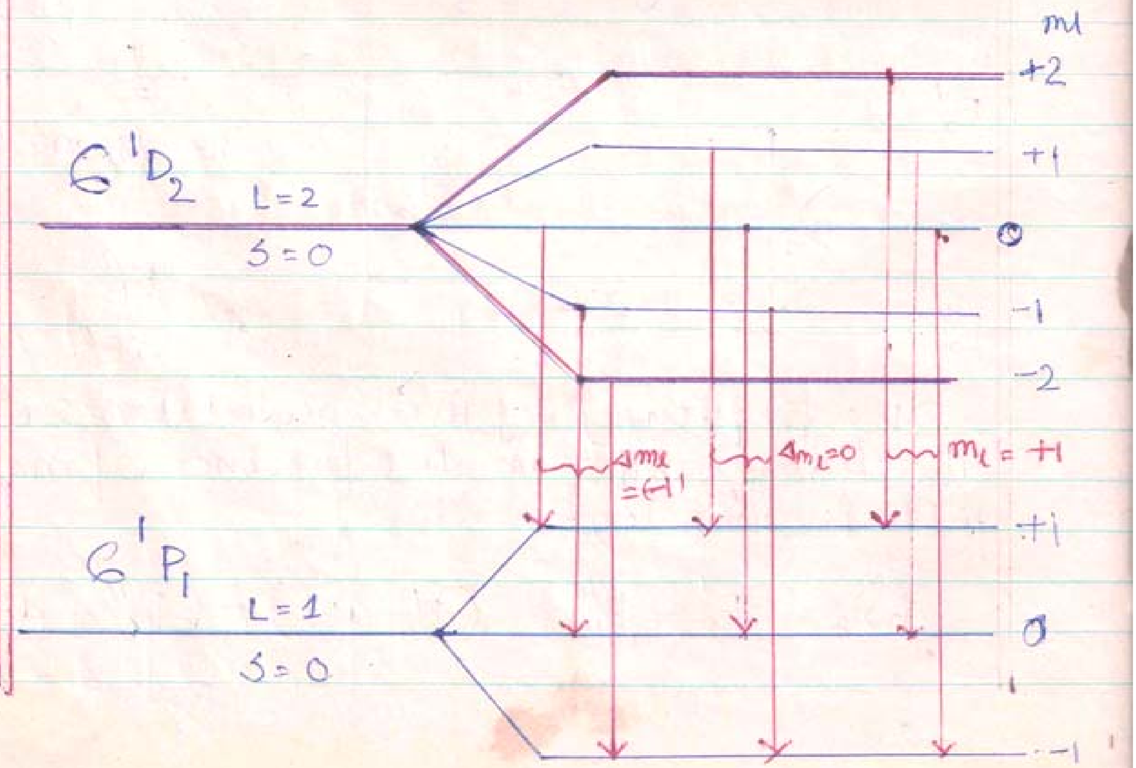
$$h\nu_B = E_B' - E_B'' = (E_0' + \Delta E') - (E_0'' + \Delta E'')$$

or $h\nu_B = (E_0' - E_0'') + (\Delta E' - \Delta E'')$

or $$\boxed{h\nu_B = h\nu_0 + \mu_B (m_l' - m_l'')B} \quad -(7)$$

Thus the frequency of the spectral lines in the presence of the magnetic field depends on the values of m_l' & m_l'' which represent the orbital magnetic quantum number of the two states respectively.

Remark: The normal Zeeman Effect is observed for spin less atoms i.e two electron atoms such as Zn or Cd. A transition for Cadmium $6^1D_2 - 6^1P_1$ is shown.



[From the R.H.S → The set of 1st three lines have $\Delta m_l = +1$; $\nu_B = \nu_0 + \mu_B(+1)B$.
For the set of 2nd three lines ; $\Delta m_l = 0$; $\nu_B = \nu_0$
For the set of last three lines ; $\Delta m_l = -1$
$\nu_B = \nu_0 + \mu_B(-1)B = \nu_0 - \mu_B B$.].

Explanation Of Anomalous Zeeman Effect:

Anomalous Zeeman Effect is more common and is found for atoms with non zero spin. Since the electron possesses a magnetic moment due to its orbital motion and a magnetic moment due to its spinning motion we have to find the resultant magnetic moment of the electron which interacts with the external magnetic field.

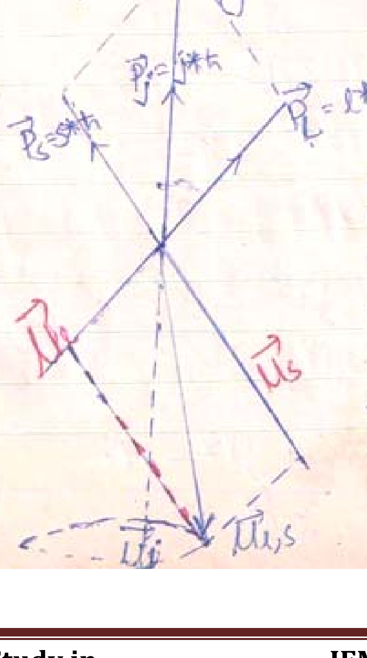
The orbital angular momentum vector $\vec{P_l} = l^* \hbar = \sqrt{l(l+1)}\hbar$ —(1)

The orbital magnetic moment vector $\vec{\mu_l} = -\frac{e}{2m}\vec{P_l} = -\mu_B \vec{l^*}$ —(2)

The spin angular momentum vector $\vec{P_s} = \vec{s^*}\hbar = \sqrt{s(s+1)}\hbar$ —(3)

The spin magnetic moment vector $\vec{\mu_s} = -2\frac{e}{2m}\vec{P_s}$

$= -2\mu_B \vec{s^*}$ —(4)



In the vector diagram $\vec{P_j} = \vec{j^*}\hbar$ represents the resultant ang. momentum of the atom, obtained by the law of parallelogram of vectors. $\vec{\mu_{l,s}}$ represents the resultant of the two magnetic moment vectors $\vec{\mu_l}$ and $\vec{\mu_s}$ by law

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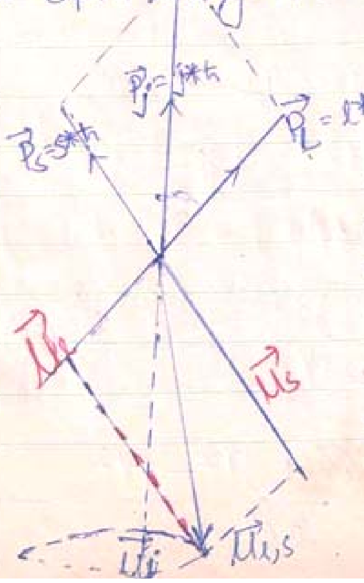

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Let $\vec{\mu_j}$ represent the resultant magnetic moment of the atom which is related to resultant angular momentum vector $\vec{P_j}$ by

$$\vec{\mu_j} = - g \frac{e}{2m} \vec{P_j} \quad \text{—— (5)}$$

Then $\vec{\mu_j}$ will be obtained from the time average of $\vec{\mu}_{l,s}$ over one complete rotation about the line representing vector $\vec{P_j}$.

If we resolve $\vec{\mu}_{l,s}$ into two mutually perpendicular components along and perpendicular to the line of $\vec{P_j}$, perpendicular components will mutually cancel out in pairs when added over a complete rotation and the sum total of all the components along the line gives $\vec{\mu_j}$.

$$\therefore \vec{\mu_j} = \mu_l \cos l^* \hat{} j^* + \mu_s \cos s^* \hat{} j^*$$

$$\text{or } \mu_j = \mu_B l^* \cos l^* \hat{} j^* + 2 \mu_B s^* \cos s^* \hat{} j^* \quad \text{—— (6)}$$

Using properties of triangle :

$$\cos l^* \hat{} j^* = \frac{l^{*2} + j^{*2} - s^{*2}}{2 l^* j^*}$$

$$\text{or } l^* \cos l^* \hat{} j^* = \frac{l^{*2} + j^{*2} - s^{*2}}{2 j^*} \longrightarrow (7)$$

Similarly

$$s^* \cos s^* \hat{} j^* = \frac{s^{*2} + j^{*2} - l^{*2}}{2 j^*} \quad \text{—— (8)}$$

But $P_j \cos\theta$ = The component of resultant angular momentum along the z-direction i.e the direction of external field.

$(\vec{P_j})_z = m_j \hbar$ where m_j is the total magnetic Quantum number which has $(2j+1)$ values from $-j$ to $+j$ in unit steps.

$$\Delta E = g \frac{eB}{2m} m_j \hbar$$

$$\text{or } \Delta E = g\, m_j\, \mu_B\, B \quad \text{—— (11)}$$

E_0' & E_0'' → energy of the two levels between which electron jumps before the magnetic field applied.

E_B' & E_B'' → Energy of the same levels after the magnetic field is applied.

$$E_B' = E_0' + \Delta E' = E_0' + g' m_j' \mu_B B.$$

$$E_B'' = E_0'' + \Delta E'' = E_0'' + g'' m_j'' \mu_B B$$

The frequency of the spectral line:-

$$h\nu_B = h\nu_0 + (g' m_j' - g'' m_j'')\, \mu_B B \rightarrow (12)$$

frequency depends on $\Delta g\, m_j = g' m_j' - g'' m_j''$

Let us consider the example of sodium D-lines which is the 1st doublet of principal series $3^2P - 3^2S$.

for 3^2P: $l = 1$, $s = \frac{1}{2}$; $j = 3/2$, $1/2$.

when $j = 3/2$; $g = 1 + \frac{3/2 \times 5/2 + 3/4 - 2}{2 \times \frac{3}{2} \times 5/2}$

$$\boxed{g = 4/3}$$

when $j = 1/2$; $\therefore g = 1 + \frac{3/4 + 3/4 - 2}{2 \times 3/4} = 1 - \frac{1}{3} = \frac{2}{3}$

For 3^2S: $l = 0$, $s = 1/2$, $j = 1/2$

$\therefore g = 1 + \frac{3/4 + 3/4}{2 \times 3/4} = 1 + 1 = 2$.